SECTION A: Attempt all questions **(55 marks)**

01.	Gathering FURPS requirements involves collecting system needs. Backend is server-side processing area of a system. A system is interconnected components working together. Server hosts services, database stores data, operating system manages hardware, SDLC guides development, API enables communication, JSON formats data, framework provides structure, UML models design efficient systems design. Answer these statements with True or False A) Backend refers to the part of a system that users directly interact with. B) FURPS include Finance, Utility, Reliability, Performance, and Security. C) A system is a collection of components that work together to achieve a goal.	**(3Marks)**
02.	Choose and circle the letter corresponding to the correct answer. (1) Which model is also known as the verification and validation model? A) Agile model B) Waterfall model C) V-Shaped model D) Prototyping model (2) Which SDLC model follows a linear and sequential approach? A) Agile model B) Waterfall model C) Prototyping model D) V-Shaped model (3) Which phase comes after system design in SDLC? A) Planning B) Maintenance C) Requirement analysis D) Implementation	**(3Marks)**
03.	Choose and circle the letter corresponding to the correct answer. (1) Which diagram shows interactions between users and a system?	**(3Marks)**

	A) Class diagram B) Use case diagram C) DFD D) Algorithm (2) Which diagram represents the system as a whole and its interaction with external entities? A) Context diagram B) Class diagram C) Decision tree D) Algorithm (3) What does UML stand for? A) Unified Machine Language B) Universal Modeling Language C) Unified Modeling Language D) User Model Language	
04.	Fill in the empty spaces by using the appropriate keywords. A) A tool used for creating diagrams such as UML is called _____. B) A(n)_____ is software used for drawing diagrams like flowcharts and network layouts. C) A(n) _____ is used to access and navigate websites on the internet.	**(3Marks)**
05.	SSADM is a systems development method used to design and analyze information systems. It focuses on a step-by-step approach using data flow diagrams, logical design, and documentation. SSADM improves accuracy, reduces errors, and ensures clear communication between analysts, developers, and users during system development. Read carefully these statements and answer them with True or False. A) SSADM stands for Structured System Analysis and Design Methods. B) The main objective of SSADM is to provide a structured approach to system development. C) SSADM focuses only on programming and coding activities.	**(3Marks)**
06.	Match the DFD elements in Column A with their meanings in Column B.	**(4Marks)**

	Answer	**Column A**	**Column B**	
	A…….	A. Process	1. Shows movement of data	
	B…….	B. Data flow	2. Represents input/output storage	
	C…….	C. Data store	3. Transforms data	
	D…….	D. External entity	4. Source or destination of data	
			5. Shows movement of elements	
07.	Complete these statements with the right keywords A) The documentation that includes tables, relationships, and schemas is called _____. B) A(n)_____documents show interactions between users and the system. C) The document that provides detailed technical requirements of a system is known as _____.			**(3Marks)**
08.	Match each programming language in Column A with its category in Column B in backend development.			**(3Marks)**
	Answer	**Column A**	**Column B**	
	1…….	1. JavaScript	A. Server-side scripting language	
	2…….	2. Ruby	B. Unified Modeling Language	
	3…….	3. PHP	C. Can be used both frontend and backend	
			D. Structured Query Language	
			E. Backend language with Rails framework	
09.	Choose and circle the letter corresponding to the correct answer.			**(3Marks)**

(1) Which requirement is related to system speed and efficiency?
   A) Performance requirements
   B) Supportability requirements
   C) Usability requirements
   D) Functionality requirements

(2) Which of the following is an example of functionality requirement?
   A) The system must recover quickly after failure
   B) The system must load web pages within 2 seconds
   C) The system must be easy to navigate the different pages
   D) The system must allow users to register accounts

(3) If a system continues working even after a failure, it is demonstrating:
   A) Performance
   B) Reliability
   C) Functionality
   D) Usability

10. Match the elements of Column A with the elements of Column B. **(3Marks)**

Answer	**Column A**	**Column B**
1……	1. Analysis phase	A. Verifying that the system meets user requirements
2……	2. Design phase	B. Installing the system in the user's environment
3…….	3.Implementation phase	C. Planning how the system will work
		D. Understanding what the system should do
		E. Building the software system

11. SSADM techniques include logical data modeling, data flow modeling, and entity behavior modeling. They help analyze, design, and structure systems by representing data, processes, and interactions clearly and systematically. **(3Marks)**

	Answer with True or False to the following statements A) Logical Data Modeling represents how data is stored and structured in a system independently of how it is physically implemented. B) Data Flow Modeling focuses on how data is physically stored in databases. C) Entity Behavior Modeling describes how entities in a system respond to events and change over time.	
12.	A database is an organized collection of data stored electronically. It allows efficient storing, retrieving, updating, and managing of information, commonly used in systems like banking, schools, and businesses. Answer with True or False to the following statements. A) Each table in a database must have the same number of columns to form relationships. B) Each table in a database stores completely unrelated information. C) Relationships in a database are used to connect data between different tables.	**(3Marks)**
13.	Fill in the empty spaces in the following statements by using FURPS. A) A software system that allows users to log in, process payments, and generate reports is demonstrating the FURPS component called _____. B) If a website loads pages quickly even when many users are online at the same time, it is optimized for _____. C) When designing an app, making it easy for users to learn and navigate improves _____.	**(3Marks)**
14.	Arrange the following steps in the correct order (**Ascending order**) for creating a use case diagram. • Define relationships • Identify the system boundary • Draw and refine the diagram • Identify the actors • Identify the use cases	**(5Marks)**

15. (3Marks)

Complete these statements by using the correct keywords. Use these keywords: **Use case Document, System Design document, Functional Specification Document, Database design Document, and Technical Specification Document.**

A) During system development, programmers rely on the _____to understand system structure, database design, and technical requirements.

B) A company wants to clearly define both functional and technical aspects of a new system. They should use the _____.

C) When a developer is using a document that explains what the system should do from a user perspective, they are referring to the _____.

16. (4Marks)

Arrange the following steps in the correct order (**Ascending order**) when you are developing a Data Flow Diagram (DFD) of a system.

- Identify data stores
- Identify external entities
- Identify data flows
- Identify processes

17. (3Marks)

Match relationship types in Column A with their uses in Column B.

Answer	**Column A**	**Column B**
1……	1.One-to-One relationship	A. A student can enroll in many courses
2……	2. One-to-Many relationship	B. Each passport belongs to one person only
3…….	3. Many-to-Many relationship	C. A teacher can teach many students in different classes

Section B: Attempt any three (3) questions **(30 marks)**

18.	A software developer is designing a Library Management System and must decide whether to use a Data Flow Diagram (DFD) or an Entity Relationship Diagram (ERD). Analyze the differences between these two diagrams by comparing their purpose, symbols, focus, level of detail, and usage.	**(10marks)**
19.	A system analyst is reviewing different components of a software project. Categorize each element into the appropriate document type: Functional Specification Document, Technical Specification Document, or Database Design Document. Elements are Security, Courses entity, System architecture, Bill payments, Session management, Customers entity, APIs, Class management, Students entity, and Student registration.	**(10marks)**
20.	A system analyst is designing a Data Flow Diagram (DFD) for a school management system. Analyze the differences between a Level 0 DFD and a Level 1 DFD by comparing them based on the following elements: definition, level of detail, processes, data flow, and purpose.	**(10marks)**
21.	An online student registration information system is a computerized platform that allows students to register for courses and manage academic records efficiently. It is designed using key system analysis tools such as a Context Diagram, which shows the system interaction with external entities like students, administrators, and registrars. A Data Flow Diagram (DFD) illustrates how registration data moves through processes. An Entity Relationship Diagram (ERD) defines relationships between entities such as students, courses, and registrations. A Use Case Diagram shows user interactions, while a Decision Table represents rules for course approval and validation. Answer the following questions given below. A) A system analyst claims that a Data Flow Diagram (DFD) is more useful than an Entity Relationship Diagram (ERD) during system analysis. Do you agree or disagree? Justify your answer. **(3marks)** B) Which is more effective for defining system boundaries: a Context Diagram or a Use Case	**(10marks)**

	Diagram? Evaluate your choice. **(2marks)** C) Evaluate whether an ERD alone is sufficient to design a complete information system. **(2marks)** D) Between a DFD and a Context Diagram, which provides a clearer understanding of system processes? Justify your evaluation. **(3marks)**	
22.	Read carefully this scenario given below and answer the questions. A banking system is a software application designed to manage customer accounts, transactions, and financial services securely. It is developed using UML (Unified Modeling Language) to visualize system structure and behavior. A class diagram defines key entities like customers, accounts, and transactions. Algorithms describe step-by-step procedures for operations such as deposits and withdrawals, while flowcharts illustrate these processes visually. A decision tree helps in making logical decisions, such as loan approval conditions. A) Evaluate whether UML diagrams are more effective than flowcharts for representing system design. Justify your answer. **(3marks)** B) Evaluate whether a UML class diagram is more effective than a decision tree for designing a banking system database structure. **(3marks)** C) A developer prefers using algorithms instead of flowcharts when designing a complex system. Critically evaluate this choice. **(4marks)**	**(10marks)**

Section C: Attempt only one (1) question **(15 marks)**

23.	A software analyst is tasked with designing a School Management Information System (SMIS) for ACB High School. To begin, he/she creates a Level 0 Data Flow Diagram (context diagram) to establish the system's boundaries and major interactions with external entities. You are requested to create a level 0 data flow diagram for this system with at least four external entities and show two interactions for each external entity, one input and one output.	**(15marks)**

24.	A library management system is a computerized system designed to manage library operations efficiently. It helps with organizing books, members, borrowing, and returning processes. The system allows users such as students and staff to search for books, borrow them, and return them within a specified time. It keeps records of book details, availability, and user information. Librarians can easily track issued books, overdue returns, and fines. The system improves accuracy, reduces manual work, and saves time. It also ensures proper cataloging of books and provides quick access to information. Design physical data model of students, books, and borrowings tables.	**(15marks)**

END OF QUESTION PAPER